

WORK & EXPERIENCE

Ergon Energy

Energy Derivatives Trader

Brisbane, QLD

Sep 2025 - Present

- Contributed to the management of a \$1.5bn hedge book across the National Electricity Market, covering physical and contractual exposure across power, gas, and carbon on OTC and exchange venues.
- Designed and researched systematic battery trading strategies, incorporating price dynamics, congestion, volatility, and operational constraints.
- Sole architect and developer of a quantitative modelling suite encompassing forward curve construction, derivative pricing engines, generation and load signal stacks, congestion and volatility models, and full trading dashboards.
- Built analytics pipelines for forward curves, spreads, and option-like payoff structures in electricity derivatives.

Senex Energy

Graduate Energy Trader

Brisbane, QLD

Nov 2024 - Sep 2025

- Built a Risk Analytics framework integrating Python & PowerBI to deliver key metrics such as VaR, CVaR, Marginal VaR.
- Learnt the market functions and portfolio balancing strategies used to trade physical energy on the STTMs & GSH.
- Created a novel way to measure the 'Macaulay duration' of a gas portfolio.

Sitesec

Systems Engineer

Gold Coast, QLD

Nov 2022 - Jul 2023

- Managed databases and ensured data integrity, accuracy, and availability for stakeholders.
- Worked with senior developers on Convolutional Neural Networks to reduce false positives at a 24% higher rate.
- Wrote custom scripts to increase firmware efficiency and reduce common router failure issues.

Eante

Software Development Intern

Brisbane, QLD

Jan 2021 - May 2021

- Developed a micro interest-earning system with visual representations to encourage saving habits.
- Gained exposure to development of explosion mapping software for an international mining project.

Nabla - Bond University

President

Gold Coast, QLD

Feb 2023 - May 2024

- Founded the Quantitative Finance Society in response to growing demand for quantitative analytics skills.
- Developed a heat map of open interest & black-scholes based price calculator using Python and Streamlit.

EDUCATION & ACCREDITATION

Australian Financial Markets Association

Financial Markets Certificate

Accredited Aug 2026

University of New South Wales

Master of Mathematics

Sydney, NSW

Feb 2025 - Nov 2027

- 80% WAM, UNSW Fintech Society
- Courses: Data Structures & Algorithms, ML & SL, Real Analysis, Measure Theory, Bayesian Inference, Advanced Stochastic Processes

Bond University

B. Actuarial Science, Major in Finance

Gold Coast, QLD

Feb 2022 - Jul 2024

- Courses: Mathematical Statistics, Econometrics, Portfolio Analysis, Risk Management, Financial Models, Game Theory

COMPETITIONS & RESEARCH

6th Place @ 2024 Susquehanna Algothon - Designed trading algorithms for a universe of 50 stocks with a sharpe of 2.14. (\$9,000 Prize Pool)
(Top 100 Globally) @ IMC Prosperity 3 - Implemented trading strategies on different asset types in a simulated universe.

SKILLS

Analytics

🐍 Python, 📊 Pandas, ⚡ Spark, 🍷 Scikit, 📉 Plotly, 🔥 PyTorch, 📈 CVXPY, 📊 Statsmodels

ADDITIONAL INFO

Interests

Acting, Gastronomy, Guitar, Piano, Tennis, NFL, NBA, Music, Fashion, Gaming

Volunteering

Certificate in Active Volunteering (III), Litter Drive, Tennis Coach, Tutoring